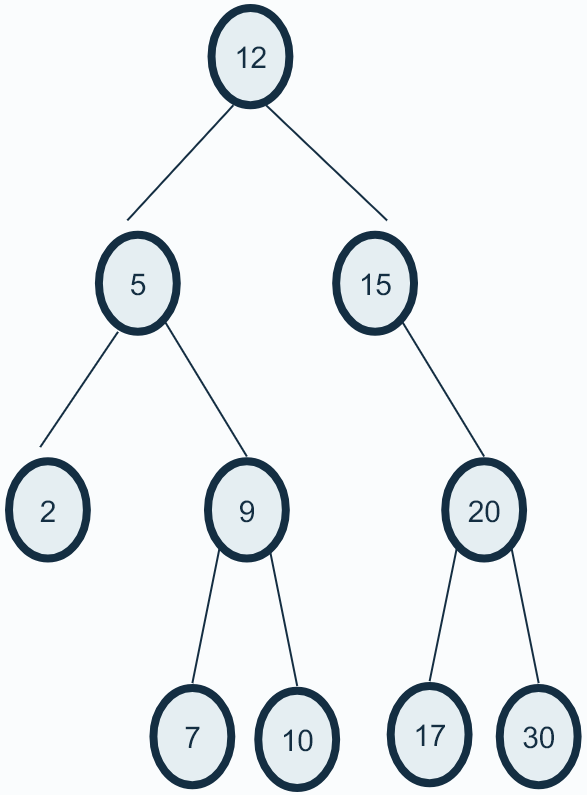
Binary Search Trees

Structure, algorithms, and the cost of height

A lesson in ordered data structures



Java: TestBST.java

Learning objectives

Explain the BST ordering rule and tree terminology.

Trace search, insertion, deletion, and traversal.

Read and implement a generic Java BST.

Relate operation cost to the height of the tree.

Why an ordered tree?

Structure	Search by key	Insert / delete
Sorted array	O(log n)	O(n) shifting
Linked list	O(n)	O(n) to locate
BST	O(h)	O(h)

A short tree can support fast search and updates.

Discussion: choosing a search structure

A course catalog supports lookup by course code.

A. The catalog rarely changes and fits in an array.
B. Courses arrive and disappear throughout the day.

Which structure would you consider for each workload?
What assumption makes BST updates fast?

Pairs: choose a structure and defend one tradeoff. 2 minutes

Trees and rooted trees

An undirected tree is a connected graph with no cycles.
A forest is an acyclic graph that may be disconnected.

A tree with n nodes has n − 1 edges and exactly one simple path between any two nodes.

Choosing a root defines parent–child relationships.

Tree terminology

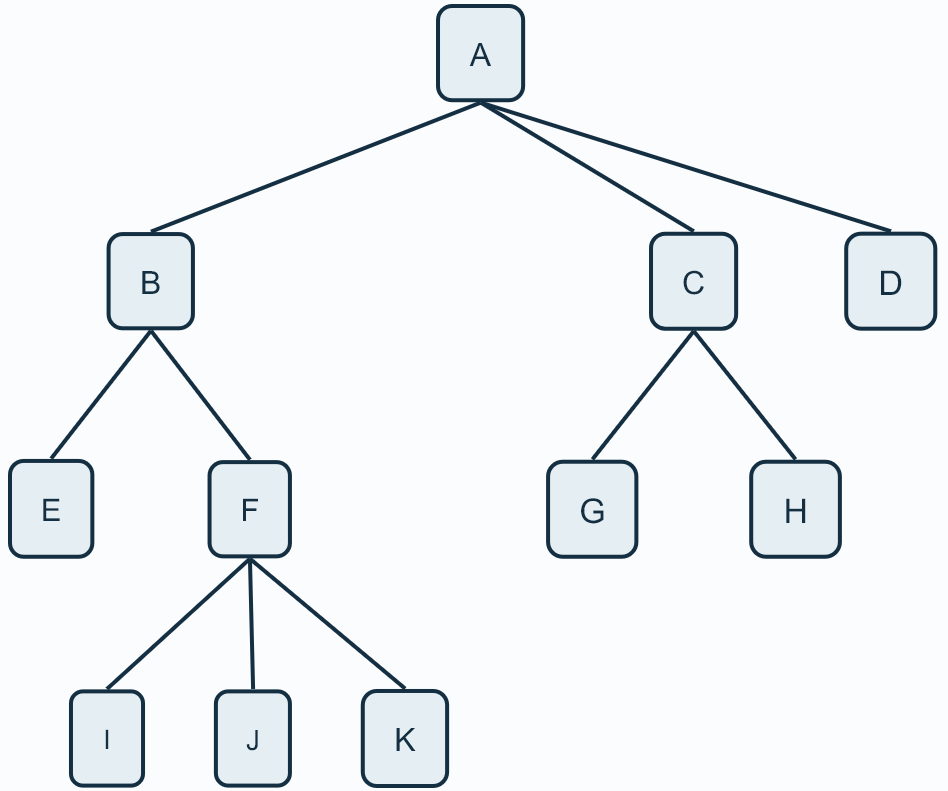
Root: A has no parent.

Leaf: E has no children.

Internal node: B has children.

Ancestors of J: F, B, A.

Subtree at F: F, I, J, K.



Binary trees and binary search trees

A binary tree gives each node at most two children, identified as left and right. Either child may be absent.

A binary search tree adds an ordering rule:
every key in the left subtree is smaller, and every key in the right subtree is larger.

Both subtrees obey the same rule.

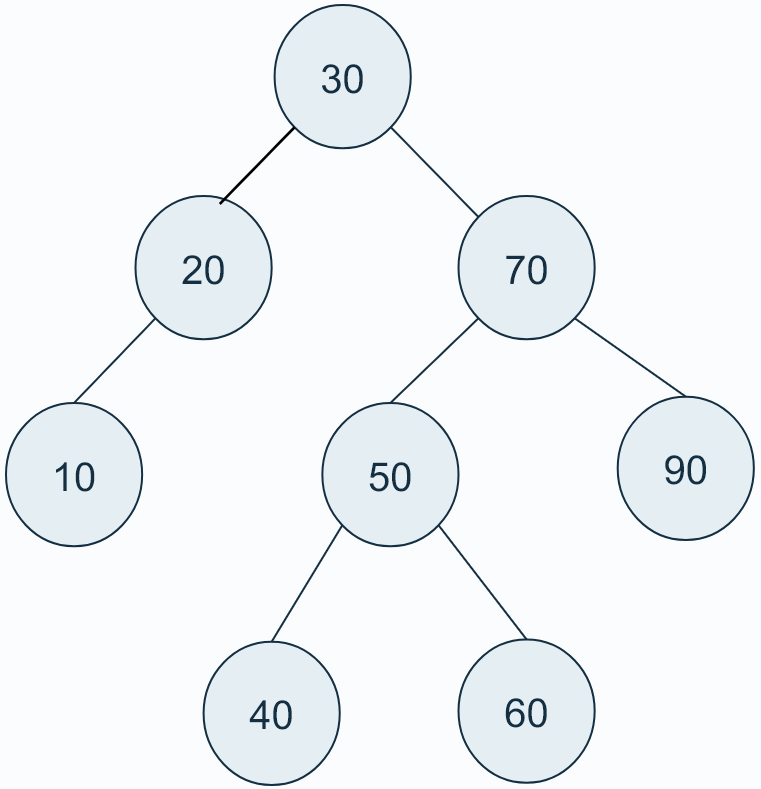
This lesson uses unique keys and rejects duplicates.

Depth, level, and height

Depth counts edges from the root.
Root 30: depth 0, level 1.
Node 40: depth 3, level 4.

Height counts occupied levels.
This tree has height 4 and 4 levels.

Empty tree: height 0.
Single node: height 1.



Java: TraversalDemo.java

Java: HeightDemo.java

Full, perfect, complete, and balanced

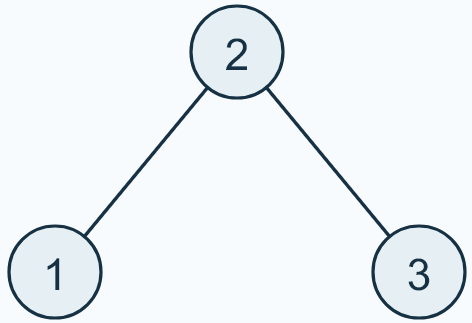
Term	Definition used in this lesson
Full (proper)	Every node has 0 or 2 children
Perfect	Full, with all leaves at the same depth
Complete	Every level fills, except possibly the last, which fills from left to right
Height-balanced	At every node, subtree heights differ by at most 1

Every perfect tree is complete. Every complete tree is height-balanced.

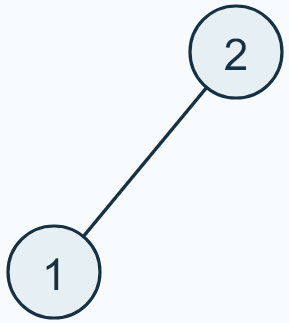
Quiz: classifying tree shapes

For A, B, and C, mark each property that holds:
full, perfect, complete, height-balanced.

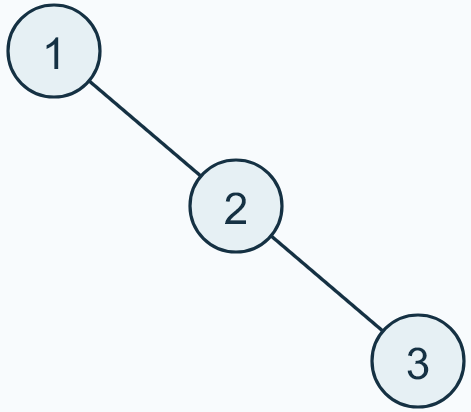
Use the definitions from the preceding table.



A



B



C

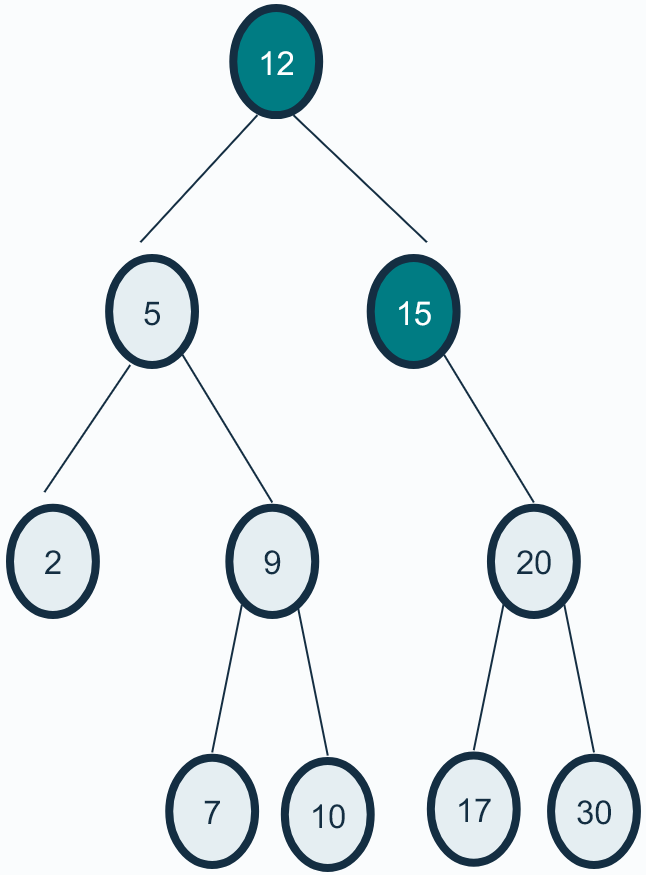
Individual response, then compare classifications. 2 minutes

The BST ordering rule applies to every descendant

At root 12:
all left keys are less than 12,
all right keys are greater than 12.

At node 15:
all right keys are greater than 15.

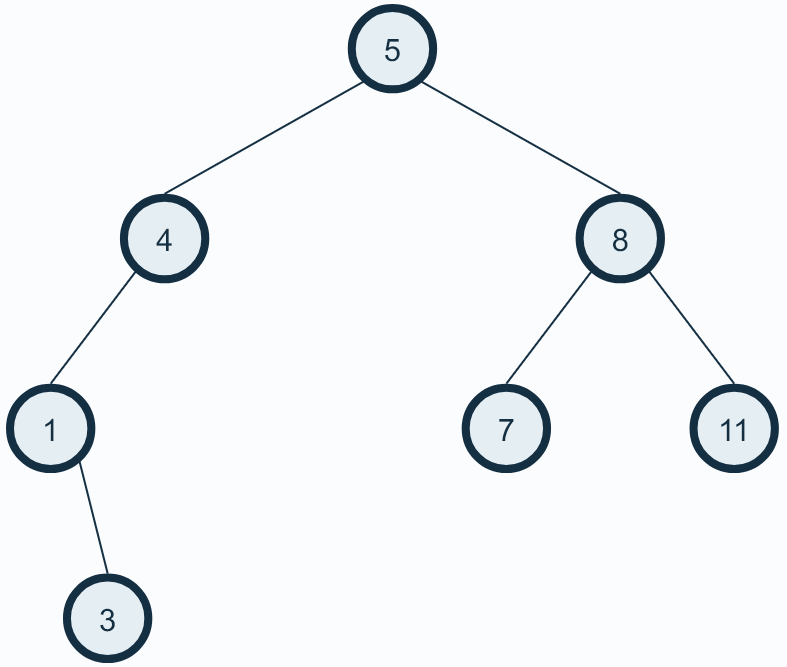
A parent–child check alone
cannot validate a whole BST.

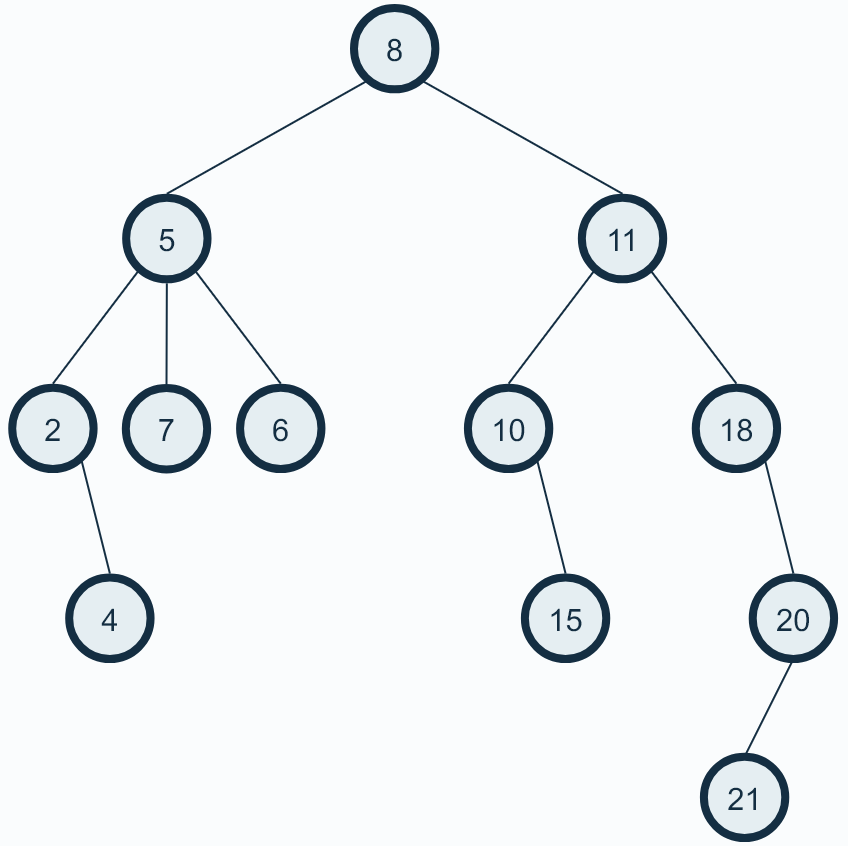


Java: TestBST.java

Java: BSTPractice.java

Checkpoint: which drawing is a valid BST?





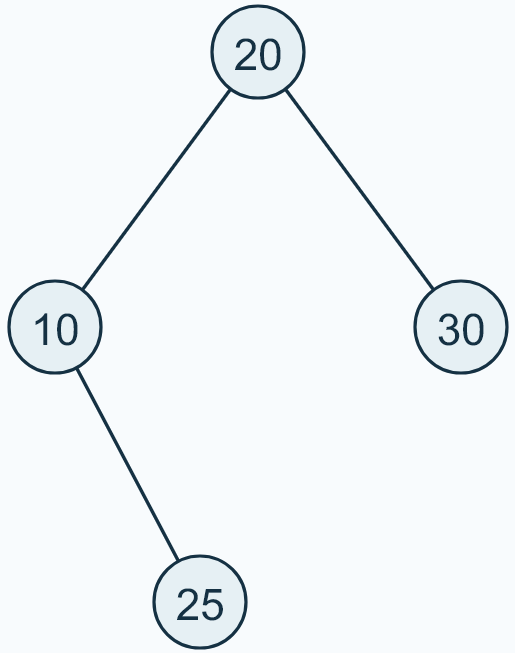
For each node, carry the lower and upper bounds from its ancestors.

Java: BSTPractice.java

Discussion: a local check can miss an error

Every left child is smaller than its parent.
Every right child is larger than its parent.

Does this drawing satisfy the BST rule?
Which ancestor constrains key 25?
What interval should contain that key?



Explain the violation using a lower and an upper bound. 2 minutes

Java: BSTPractice.java

A linked representation in Java

```
class Node<E> {
    E key;
    Node<E> left, right;

    Node(E key) { this.key = key; }
}

class BST<E extends Comparable<? super E>> {
    private Node<E> root;
    private int size;
    // Methods follow.
}
```

A missing child uses null. The tree stores a root reference and a size.

Java: Node.java

Java: BinarySearchTree.java

The Tree interface and BST implementation

```
interface Tree<E> extends Iterable<E> {
    boolean contains(E key);
    boolean insert(E key);
    boolean delete(E key);
    int size();
}
```

BST<E> implements Tree<E>.
insert returns false for a duplicate key.
delete returns false when the key is absent.

Use compareTo consistently for less than, greater than, and equality.

Java: Tree.java

Java: OrderingEqualityDemo.java

Quiz: what counts as a duplicate?

```
BST<BigDecimal> t = new BST<>();
boolean a = t.insert(new BigDecimal("1.0"));
boolean b = t.insert(new BigDecimal("1.00"));
int n = t.size();
```

Predict a, b, and n.

BigDecimal compares these values as numerically equal,
but equals also considers their scales.
Which equality rule does this BST use?

Predict before opening the Java example. 90 seconds

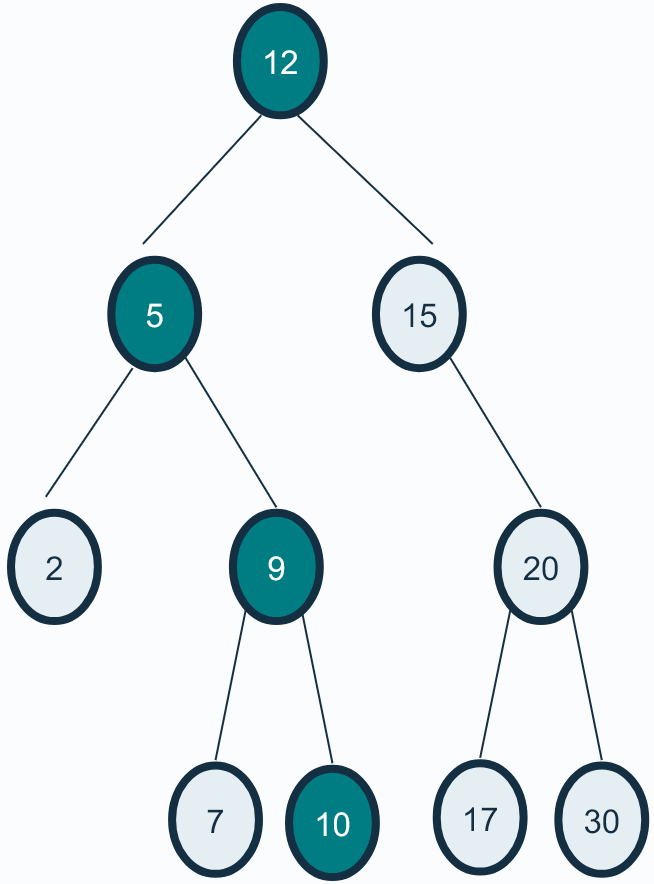
Java: OrderingEqualityDemo.java

Search follows one path

Search for 10

10 < 12: go left.
10 > 5: go right.
10 > 9: go right.
10 = 10: found.

A null child ends an unsuccessful search.



Java: SearchDemo.java

Iterative search

```
public boolean contains(E key) {
    Objects.requireNonNull(key);
    Node<E> x = root;
    while (x != null) {
        int cmp = key.compareTo(x.key);
        if (cmp < 0) x = x.left;
        else if (cmp > 0) x = x.right;
        else return true;
    }
    return false;
}
```

Time O(h). Extra space O(1). Import java.util.Objects.

Java: BinarySearchTree.java

Java: SearchDemo.java

Recursive search

```
private boolean contains(Node<E> x, E key) {
    if (x == null) return false;
    int cmp = key.compareTo(x.key);
    if (cmp < 0) return contains(x.left, key);
    if (cmp > 0) return contains(x.right, key);
    return true;
}
```

The public method checks the key, then calls contains(root, key).

Time O(h). Call stack O(h).

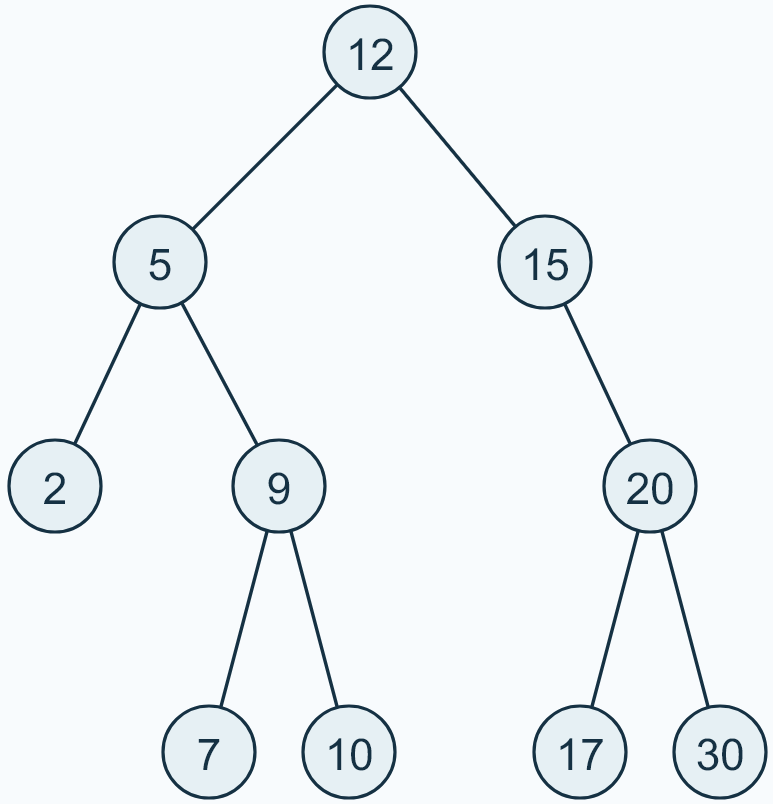
Java: BinarySearchTree.java

Java: SearchDemo.java

Quiz: an unsuccessful search

Search for key 8 in this tree.

1. List each visited key in order.
2. Which missing child ends the search?
3. How many compareTo calls occur?
4. Compare iterative and recursive auxiliary space.



Trace independently before checking with a partner. 2 minutes

Java: SearchDemo.java

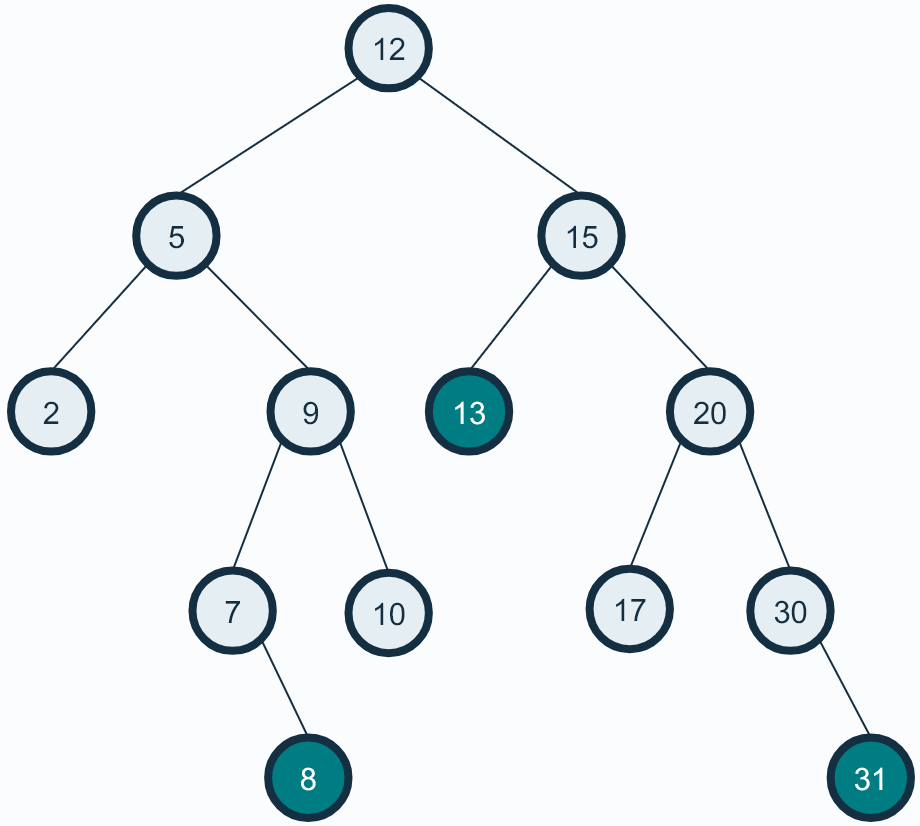
Insertion attaches a new leaf

Search for the new key.

If an equal key exists, return false.

Otherwise, attach a new node
at the first missing child link.

An empty tree gets a new root.



Java: InsertionDemo.java

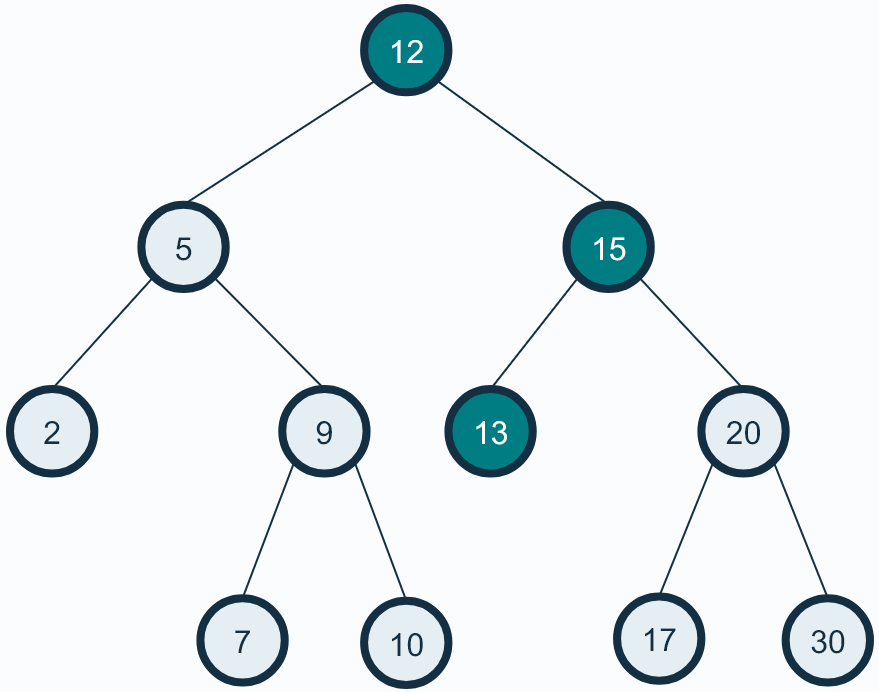
Worked insertion: key 13

Compare with 12: 13 is larger.

Compare with 15: 13 is smaller.

The left link of 15 is empty.
Attach node 13 there.

The BST ordering rule still holds.



Java: InsertionDemo.java

Iterative insertion

```
public boolean insert(E key) {
    Objects.requireNonNull(key);
    Node<E> parent = null, x = root;
    int cmp = 0;
    while (x != null) {
        parent = x;
        cmp = key.compareTo(x.key);
        if (cmp == 0) return false;
        x = cmp < 0 ? x.left : x.right;
    }
    Node<E> node = new Node<>(key);
    if (parent == null) root = node;
    else if (cmp < 0) parent.left = node;
    else parent.right = node;
    size++;
    return true;
}
```

Java: BinarySearchTree.java

Java: InsertionDemo.java

Discussion: insertion order changes the shape

Build two ordinary BSTs from the same seven keys.

A: 1, 2, 3, 4, 5, 6, 7
B: 4, 2, 6, 1, 3, 5, 7

Sketch each tree and count its height in levels.
Compare searching for 7 and performing a full traversal.
What does the common inorder sequence tell you?

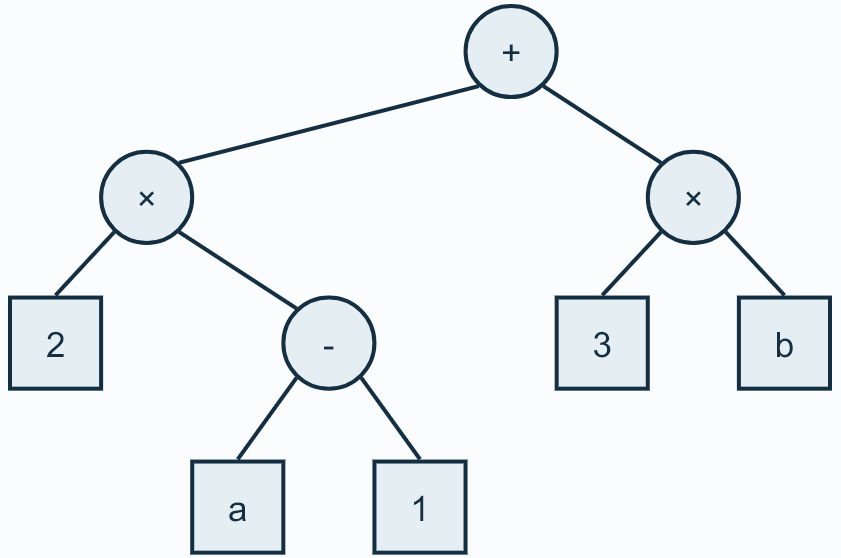
Groups of two: one sketch each, then explain the costs. 3 minutes

An expression tree is a binary tree

Expression:
2 × (a − 1) + 3 × b

Internal nodes hold operators.
Leaves hold operands.

The shape captures evaluation order rather than key ordering.



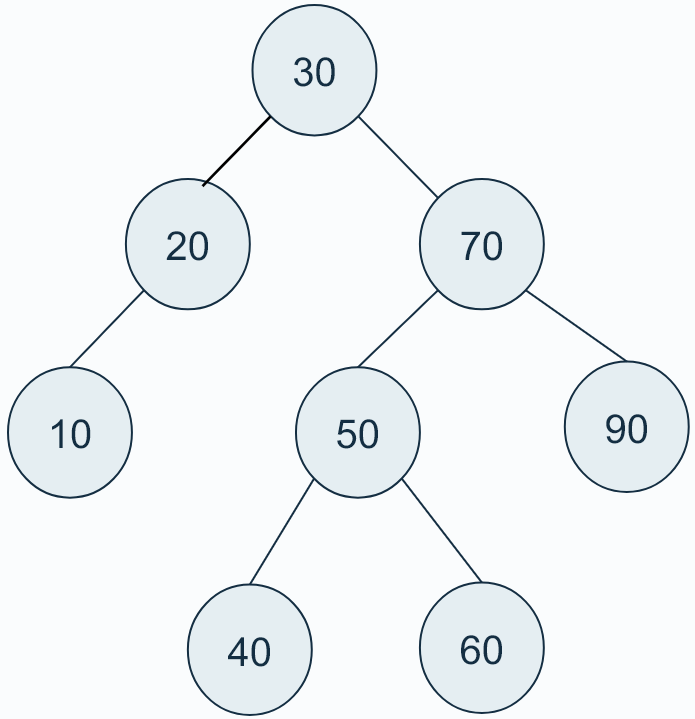
Java: ExpressionTreeDemo.java

Traversal visits every node once

Depth-first search follows a branch before returning to other branches.

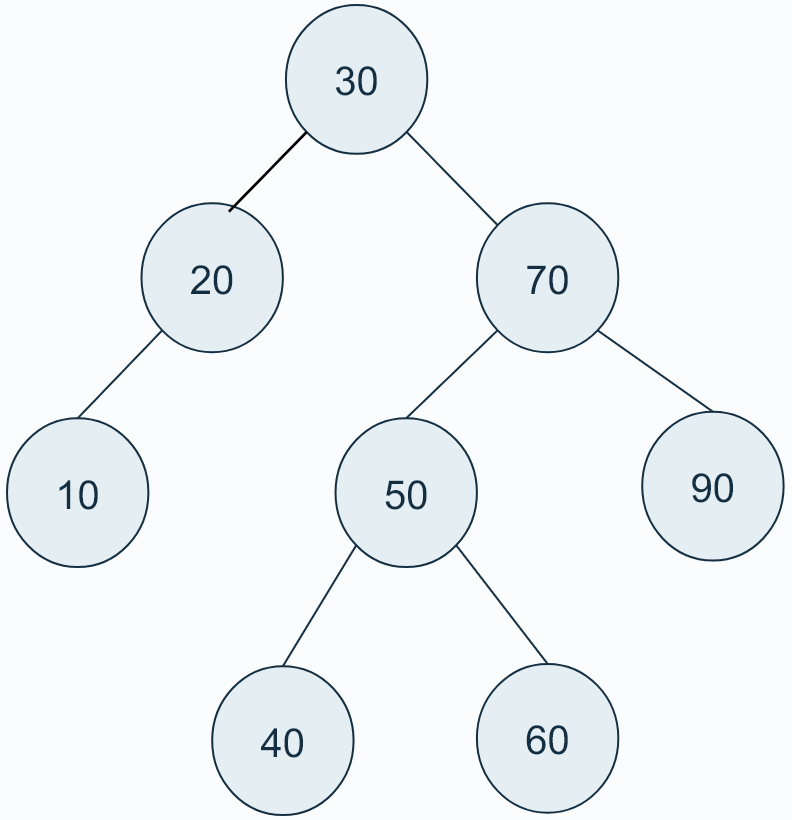
Preorder: node, left, right.
Inorder: left, node, right.
Postorder: left, right, node.

Breadth-first search visits nodes one level at a time.



Java: TraversalDemo.java

Traversal orders on the same tree



Preorder
30, 20, 10, 70, 50, 40, 60, 90

Inorder
10, 20, 30, 40, 50, 60, 70, 90

Postorder
10, 20, 40, 60, 50, 90, 70, 30

Inorder traversal of a BST lists its keys in sorted order.

Java: TraversalDemo.java

Recursive depth-first traversal

```
void inorder(Node<E> x) {
    if (x == null) return;
    inorder(x.left);
    visit(x.key);
    inorder(x.right);
}
```

Preorder moves visit before both recursive calls.
Postorder moves visit after both recursive calls.

visit is the action to perform on a key, such as printing it.

All three take O(n) time and O(h) stack space.

Java: BinarySearchTree.java Java: TraversalDemo.java

Breadth-first traversal uses a queue

```
void levelOrder() {
    if (root == null) return;
    Queue<Node<E>> q = new ArrayDeque<>();
    q.add(root);
    while (!q.isEmpty()) {
        Node<E> x = q.remove();
        visit(x.key);
        if (x.left != null) q.add(x.left);
        if (x.right != null) q.add(x.right);
    }
}
```

Time O(n). Queue space O(w), where w is the maximum level width.

Java: BinarySearchTree.java

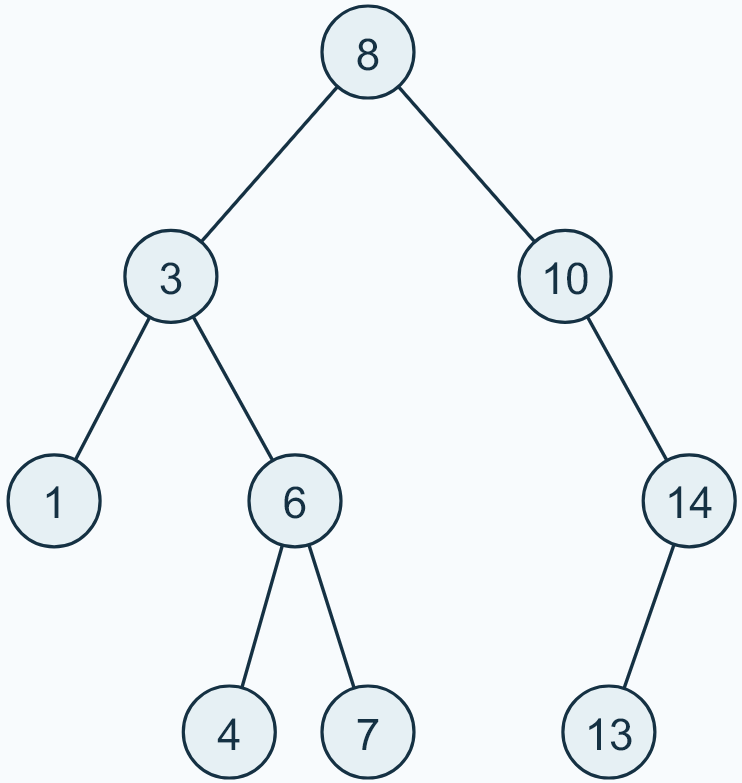
Java: TraversalDemo.java

Quiz: inorder and breadth-first order

For this new tree:

1. Write the inorder sequence.
2. Write the breadth-first sequence.
3. Starting with [8], what is in
   the queue after removing
   and processing 8, then 3?

Show the queue front first.



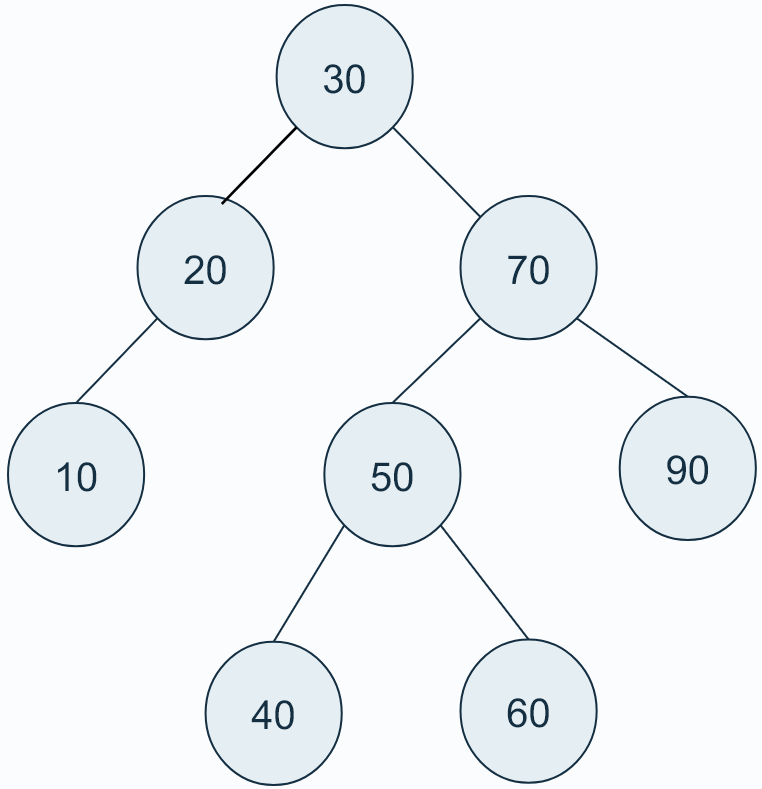
Keep a written queue trace. 3 minutes

Checkpoint: traversal and successor

Write the inorder, preorder, and postorder sequences.

Which key follows 30 in inorder traversal?

Explain why that key is the inorder successor.



Java: TraversalDemo.java

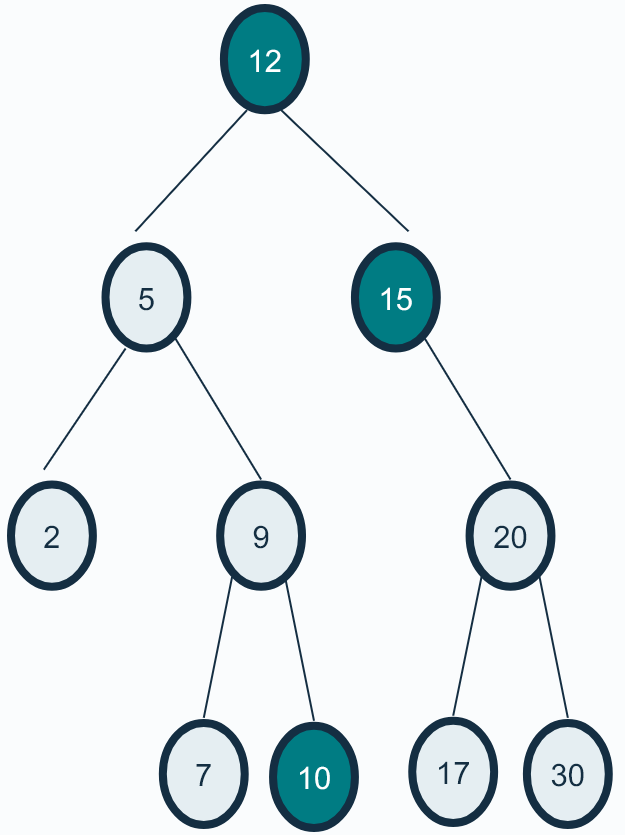
Predecessor and successor

For root 12:

Predecessor = 10
Largest key in the left subtree.

Successor = 15
Smallest key in the right subtree.

The maximum follows right links.
The minimum follows left links.



Java: BinarySearchTree.java

Java: TestBST.java

Deletion has three structural cases

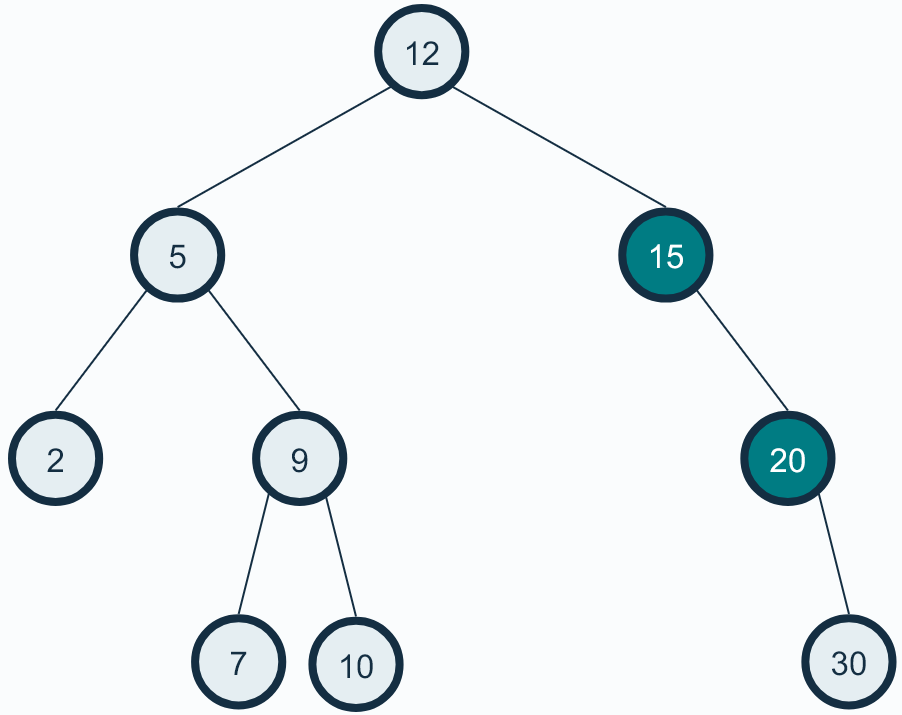
Children	Repair after finding the node
0	Remove the parent’s child link
1	Connect the parent directly to the child
2	Copy the successor key, then remove the successor from the right subtree

Deleting the root can change root. An absent key leaves size unchanged.

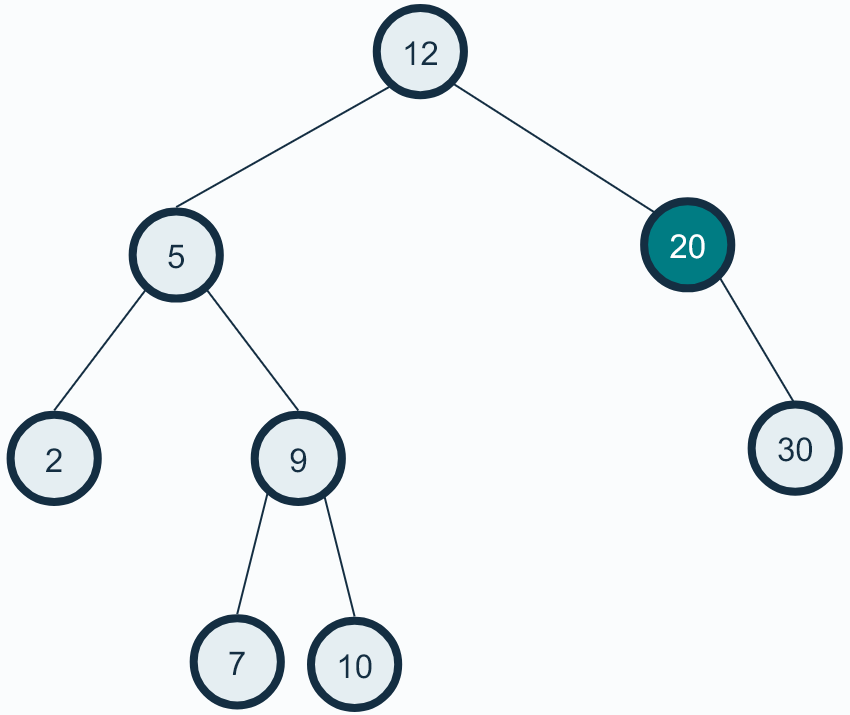
Java: DeletionDemo.java

One child: deletion reconnects the subtree

Before: delete 15



After: 12 links to 20



A leaf uses the same reconnection idea with a null replacement.

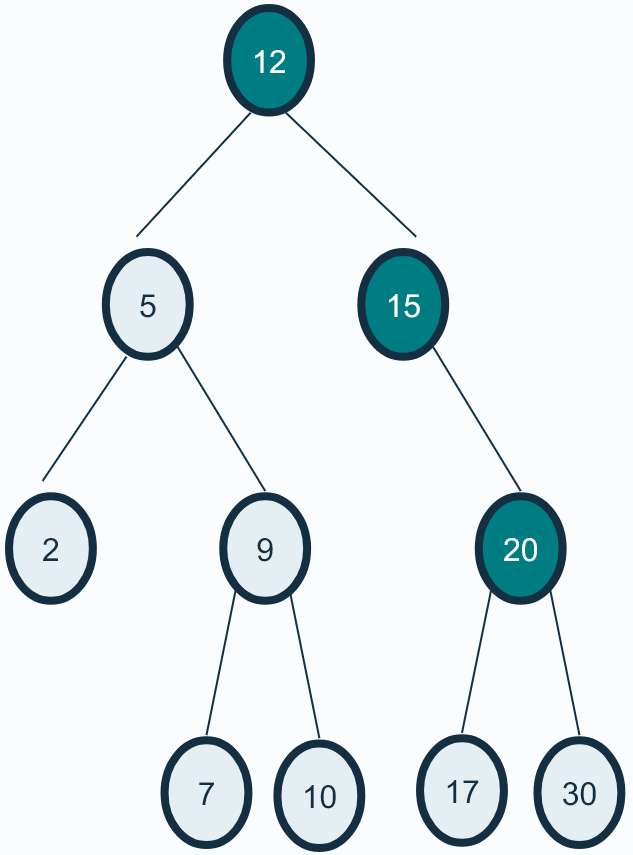
Java: DeletionDemo.java

Two children: deletion uses a neighboring key

Delete 12 using its successor.

1. Find the minimum in the right subtree: key 15.
2. Copy 15 into the root.
3. Remove the original node 15, reconnecting its child 20.

The predecessor is an equally valid alternative.

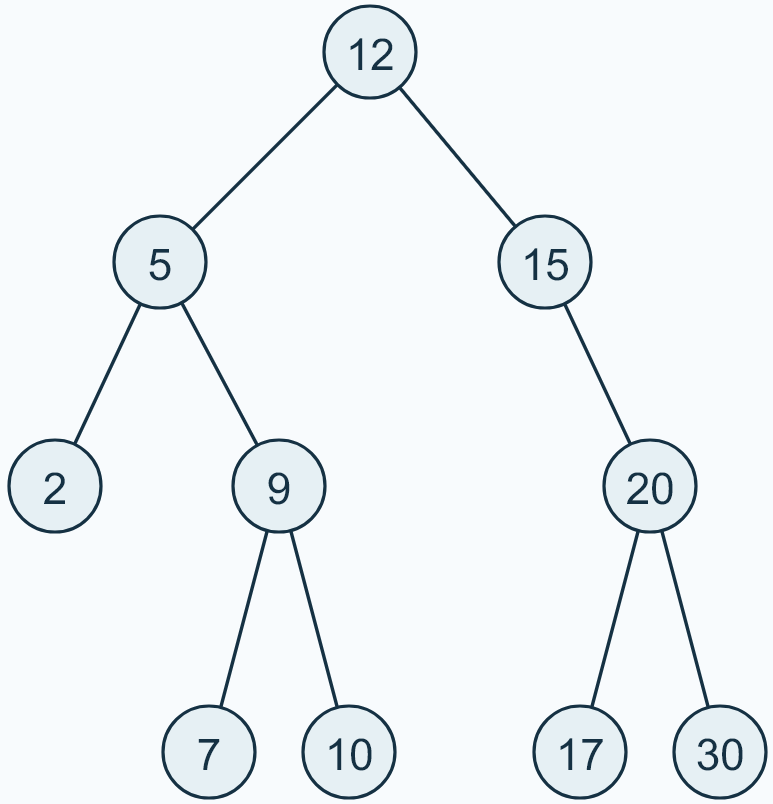


Java: DeletionDemo.java

Quiz: deleting a root with two children

Use the tree shown, independently
of earlier deletion examples.

Delete 12 using its successor.
1. Which key becomes the root?
2. Which original node disappears?
3. Where does subtree 20 reconnect?
4. Which keys remain in inorder?



Draw the changed links before writing the output. 3 minutes

Java: DeletionDemo.java

Recursive deletion preserves the BST rule

```
private Node<E> delete(Node<E> x, E key) {
    if (x == null) return null;
    int cmp = key.compareTo(x.key);
    if (cmp < 0) x.left = delete(x.left, key);
    else if (cmp > 0) x.right = delete(x.right, key);
    else {
        if (x.left == null) return x.right;
        if (x.right == null) return x.left;
        x.key = min(x.right).key;
        x.right = deleteMin(x.right);
    }
    return x;
}
```

Each call reconnects the returned subtree root to its parent.

Java: BinarySearchTree.java Java: DeletionDemo.java

Successor helpers

```
private Node<E> min(Node<E> x) {
    while (x.left != null) x = x.left;
    return x;
}

private Node<E> deleteMin(Node<E> x) {
    if (x.left == null) return x.right;
    x.left = deleteMin(x.left);
    return x;
}
```

The successor has no left child. It may have a right child to reconnect.

Java: BinarySearchTree.java

The public deletion method

```
public boolean delete(E key) {
    Objects.requireNonNull(key);
    if (!contains(key)) return false;
    root = delete(root, key);
    size--;
    return true;
}
```

The preliminary search keeps size bookkeeping simple.
A later optimization can combine search and deletion.
Both versions take O(h) time.

Java: BinarySearchTree.java

Java: DeletionDemo.java

Discussion: a returned subtree matters

```
// Inside delete, when key < x.key:
delete(x.left, key);
return x;
```

Suppose root 12 has a single left leaf, key 5.
The recursive call returns null when deleting that leaf.

Why does this fragment leave 5 connected?
Which assignment repairs the link?
Why does the public method assign root as well?

Explain the returned reference, then write the repaired line. 2 minutes

Java: BinarySearchTree.java

An inorder iterator

An iterator exposes hasNext() and next().

A lazy inorder iterator keeps a stack of ancestors.
Initially, push the path from root to the leftmost node.

On next(), pop a node and return its key.
Push the leftmost path in that node's right subtree.

Across n calls, total time is O(n) with O(h) stack space.

Exercise: handle an exhausted iterator with NoSuchElementException.

Java: BinarySearchTree.java Java: IteratorDemo.java

Discussion: the iterator uses a stack

A lazy inorder iterator stores the path to the next key.

1. Why can its stack use O(h) space?
2. Can one next() call take O(h) time?
3. Why do n successful calls take O(n) total time?
4. What should next() do after exhaustion?

Explain the accounting with one push and one pop per node. 2 minutes

Java: IteratorDemo.java

Height determines operation cost

Operation	General BST	If height is O(log n)
Search, insert, delete	O(h)	O(log n)
Minimum, maximum	O(h)	O(log n)
Full traversal	O(n)	O(n)

Sorted insertion can create a chain with height n.
A balanced search tree controls height as the data changes.

Java: HeightDemo.java

Traversal cost and search cost

A full traversal visits all n nodes: O(n) time.

Search follows one root-to-node path: O(h) time.

A short tree and a chain have the same traversal cost,
but very different search paths.

Discuss: when must an algorithm visit every node?

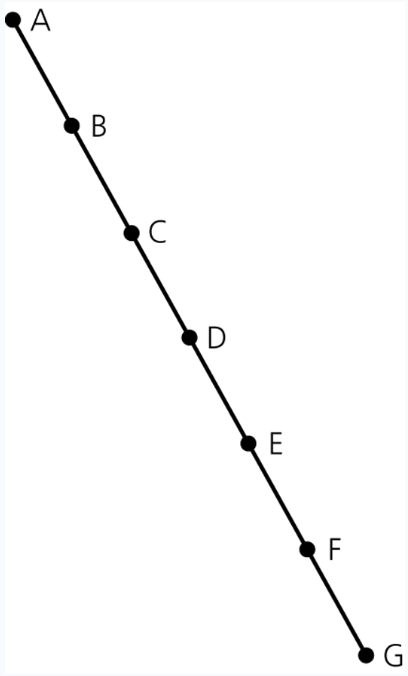
Java: TraversalDemo.java

A seven-node chain

With height counted in levels:

7 nodes form a chain of height 7.
There are 7 levels.

Searching to the deepest node visits 7 nodes: h comparisons.



Maximum-height tree with 7 nodes

Diagram: Pearson Addison-Wesley, 2011 (supplied source)

Java: HeightDemo.java

Height bounds with a node count

For height h ≥ 0, maximum nodes:

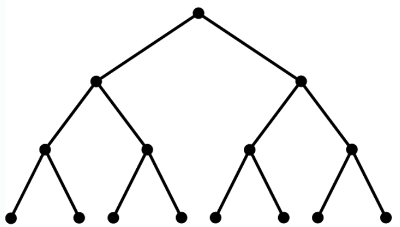
2^h − 1

At level L ≥ 1, maximum nodes:

2^(L − 1)

For n ≥ 1 nodes, height satisfies:

ceil(log2(n + 1)) ≤ h ≤ n



Level	Number of nodes at this level	Number of nodes at this and previous levels
1	$1 = 2^0$	$1 = 2^1 - 1$
2	$2 = 2^1$	$3 = 2^2 - 1$
3	$4 = 2^2$	$7 = 2^3 - 1$
4	$8 = 2^3$	$15 = 2^4 - 1$
.	.	.
.	.	.
.	.	.
h	2^{h-1}	$2^h - 1$

A perfect tree attains the maximum node count for its height.

Java: HeightDemo.java

Quiz: height and levels

Use level-count height: empty 0, leaf 1.

1. For 31 nodes, what are the minimum and maximum possible BST heights?
2. A tree has height 4. How many levels does it have, and what is its maximum node count?
3. Does a minimum-height BST have to be complete? Give a reason or a counterexample.

Show the formulas and explain the final claim. 3 minutes

Java: HeightDemo.java

Treesort

Insert the input keys into a BST.
Then emit keys in inorder.

Expected time with random insertion order: O(n log n).
Worst case for an unbalanced tree: $O(n^2)$.

A set-style BST discards duplicates. A sorting implementation must retain equal keys, for example with a count per node.

Java: TreeSort.java

Discussion: treesort must keep duplicates

Input: 4, 2, 4, 1, 2

A student inserts every value into our set-style BST
and then emits an inorder traversal.

Which output does the student get?
What output must a sorting algorithm return?
How could a count at each node preserve every item?

Compare the number of input and output items. 2 minutes

Java: TreeSort.java

Review problems

1. What is the maximum node count with 8 levels?

2. For 2011 nodes, what are the minimum and maximum numbers of levels?

3. With exactly 10 levels, what are the minimum and maximum numbers of leaves?

4. Why can sorted insertion make a BST slow?

Exit quiz: explain the invariant

1. Why can a parent-child comparison miss a BST error?

2. A node has two children. Why does its successor have at most one child?

3. A BST contains n keys. Does that alone guarantee O(log n) search? State the missing condition.

4. Can changing a stored key break the BST rule even when no links change?

Write a short explanation for each answer. 3 minutes

Further practice

Implement iterative inorder traversal with an explicit stack.

Compute height using a queue and one level at a time.

Validate a BST using lower and upper bounds.

Compare sorted insertion with random insertion.

Explain how changing a stored key could break ordering.

A BST’s ordering rule gives the algorithm. Its shape determines the cost.

Java: BSTPractice.java Java: HeightDemo.java Java: IteratorDemo.java